

American Put Option Pricing

Week 5, Part C

Reasoning behind use of non-linear models :-

A linear model is insufficient for pricing American put options because the relationship between option price and its input variables is nonlinear by nature. Variables such as stock price, time to maturity, volatility, and interest rate interact in complex ways that cannot be captured by a straight-line relationship. In particular, American put options allow early exercise, which introduces a free boundary problem and makes the pricing surface more complicated than standard European options. The option value does not change uniformly with changes in stock price or volatility. Near the strike price, small changes in inputs can lead to disproportionately large changes in option value. A linear model may capture broad trends but will fail to represent these curved and discontinuous relationships accurately. Therefore, a nonlinear model such as a neural network is more suitable for approximating the option pricing surface.

Input features to be used in Week 6:-

1. **Stock Price (S)** — measured in currency units (e.g. dollars)
 2. **Strike Price (K)** — measured in currency units, same as stock price
 3. **Time to Maturity (T)** — measured in years
 4. **Volatility (σ)** — annualized decimal value (for example 0.2 for 20%)
 5. **Risk-Free Interest Rate (r)** — annualized decimal value (for example 0.05 for 5%)
- These units match the binomial option pricing model made earlier so that the neural network receives inputs consistent with the pricing calculations.

Sanity Check for Week 6 :-

I will first check that the predicted option price is **non negative**, since an option cannot have negative value in practice. I will also check **monotonicity with respect to stock price**, meaning that for a put option, as the stock price increases, the option price should generally decrease. If the model violates either of these properties, it would suggest that the neural network has learned unrealistic behavior despite fitting the training data.